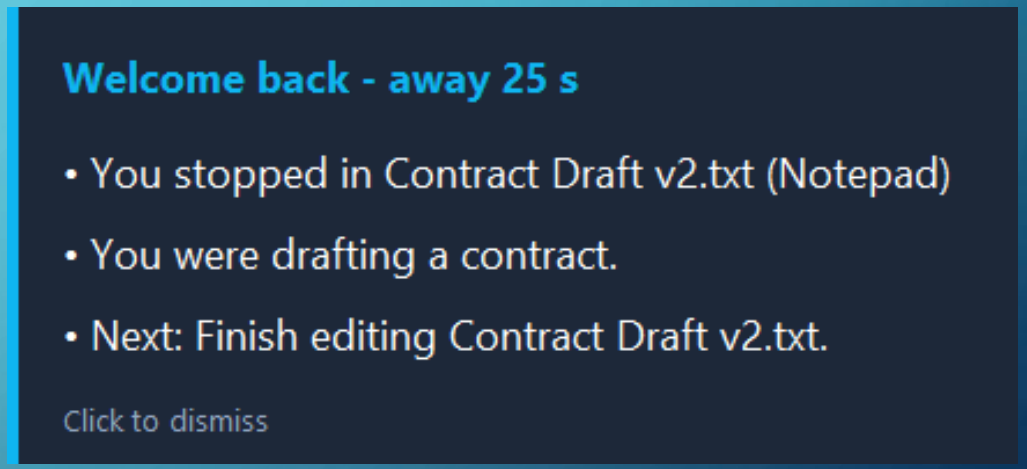
ASUS UGEN AI LEAGUE · BATTLEFIELD LIGHTNING · WORKPLACE AI · ONSITE

RECALL PLUG

RECALL PLUG SUMMARIZES YOUR INTERRUPTED WORK

SO YOU CAN EFFORTLESSLY PICK UP WHERE YOU LEFT OFF.



BY 蘇靖飛: https://github.com/geoffreysugs/recallplug

Interruptions waste your time

You’re in the middle of work, halfway through a document, when your manager suddenly calls you into an unscheduled meeting. An hour later, you can’t recall where you left off. After clicking to every open window, you finally piece together what you were doing and get back to work. The research shows that getting back to an interrupted task takes about 25 minutes on average.

What exists today

Option	How it works	Downsides
Manual: recent files, notes	Read the notes again to recall	No help when you are interrupted suddenly
Windows Recall	Screenshots every few seconds, searchable	Needs Copilot and PC to stores your screenshot
Screenpipe-style tools	Continuous local screen recording	Waste storage because it records everything you saw and said

Who meets this problem: project managers, sales, procurement and admin staff, or anyone with several documents open who is pulled away by meetings and calls.

Solution: Recall Plug, a local AI assistant on a USB accelerator

Working in a company inevitably involves business secrets. When choosing an AI that records your work, there are two concerns: it must not upload your data to the cloud, and if it runs locally instead, the workload is limited by your computer's hardware and may even slow down your actual work.

Recall Plug is designed to address these two concerns. It runs AI locally on ASUS UGen300, a USB accelerator, so an ordinary office PC does not have to be replaced or burdened with AI workloads. While you are away, it prepares a three-line recap: where you stopped, what you were doing, and what to do next. The moment you return, the recap appears at the bottom right of your screen. Recall Plug stores nothing on the UGen300: the models stay on your PC and are loaded into the device at each launch.

Current status: the prototype reads window titles and runs a local model on a laptop. The UGen300 version, reading the screen, and the ask panel are targets for Stage II.

Advantages Why a local AI on a USB accelerator.

Advantage	What it means	How we implemented it
Works offline	After downloading a one-time model, no network is needed.	The app keeps the models in its own folder; the download starts at first launch, with your consent.
No per-use token fees	Every request runs on your PC and UGen300, so there is no cloud charge required for each use, unlike cloud AI assistants.	Models run locally; UGen300 is a one-time hardware purchase.
Private by design	Only the window you are opening is readable by AI, and the image is deleted right after analysis, only a short text note is kept, and nothing is uploaded.	Via the pause key, capture indicator and block list, the network audit in Stage II will check it.
Light on your PC	The heavy AI work runs on UGen300, so an ordinary office PC does not have to be replaced.	Models are loaded into the device at each launch; on-device numbers come in Stage II.

The only network use is the one-time model download. These are design goals, not yet verified on the device.

Tools

Layer	Tool	Why
Language	Python 3.12	Free and easy to use
Local runtime	Ollama	Runs models locally; offline after the first download; no per-use token fees
Text model	Qwen2.5:1.5b, about 1.5 billion parameters	A small model(in UGen300 datasheet)
Vision model	Qwen2-VL-2B (planned)	Describes a screen as text (in the UGen300 datasheet); tested by hand with a stand-in model
Accelerator	ASUS UGen300 over USB(planned)	Runs the AI models and stores no data; models are loaded at each launch
Console	Streamlit(planned)	An English dashboard: pins, ask panel, save your memory

Functions

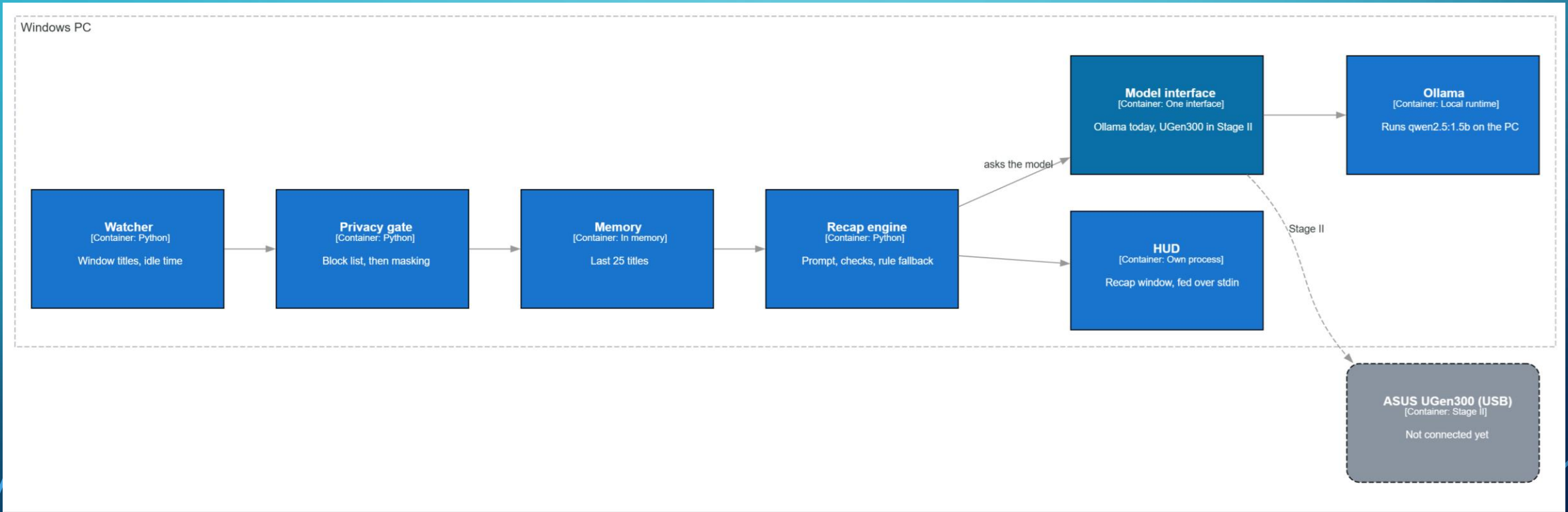
Functions	How it works	Why it helps
1. Prepare before you return	The recap is written while you work and while you are away, not when you return..	The model takes a few seconds, but you never wait for it: the recap is already written when you return.
2. Rules for facts, the model for reasoning	Rules write line 1. The model writes lines 2–3, which are checked against your recent windows; a failed line is replaced by a rule.	Line 1 is written by rules from the window title, the model only adds a guess that has been checked
3. Look briefly, keep only text(planned)	After 10 s without input, it looks at the window you are in, and writes a short text note then deletes the image at once. It checks again every 10 s and skips the note if nothing changed.	Unlike Windows Recall, no screenshot is stored, only a short text note is kept.
4. Answers you can check (planned)	You can ask about your past work; each answer uses only saved recaps and says which one.	Answers come only from saved recaps, so you can check every one.

Methodology

Principle	What we do	Why
1. Decide from experiments	We test a design choice, keep what works and drop what does not.	The design follows evidence, not guesses.
2. Label every number	Each figure is marked MEASURED, PROJECTED, ASSUMPTION or TARGET. Nothing planned is shown as done.	Readers can tell what is proven from what is expected.
3. Protect first, then add	Privacy protections come before screen-reading features; without them, the feature stays off.	The tool never collects more than its protections allow.
4. Make results reproducible	Each benchmark and check runs from one command.	Anyone can repeat the numbers.

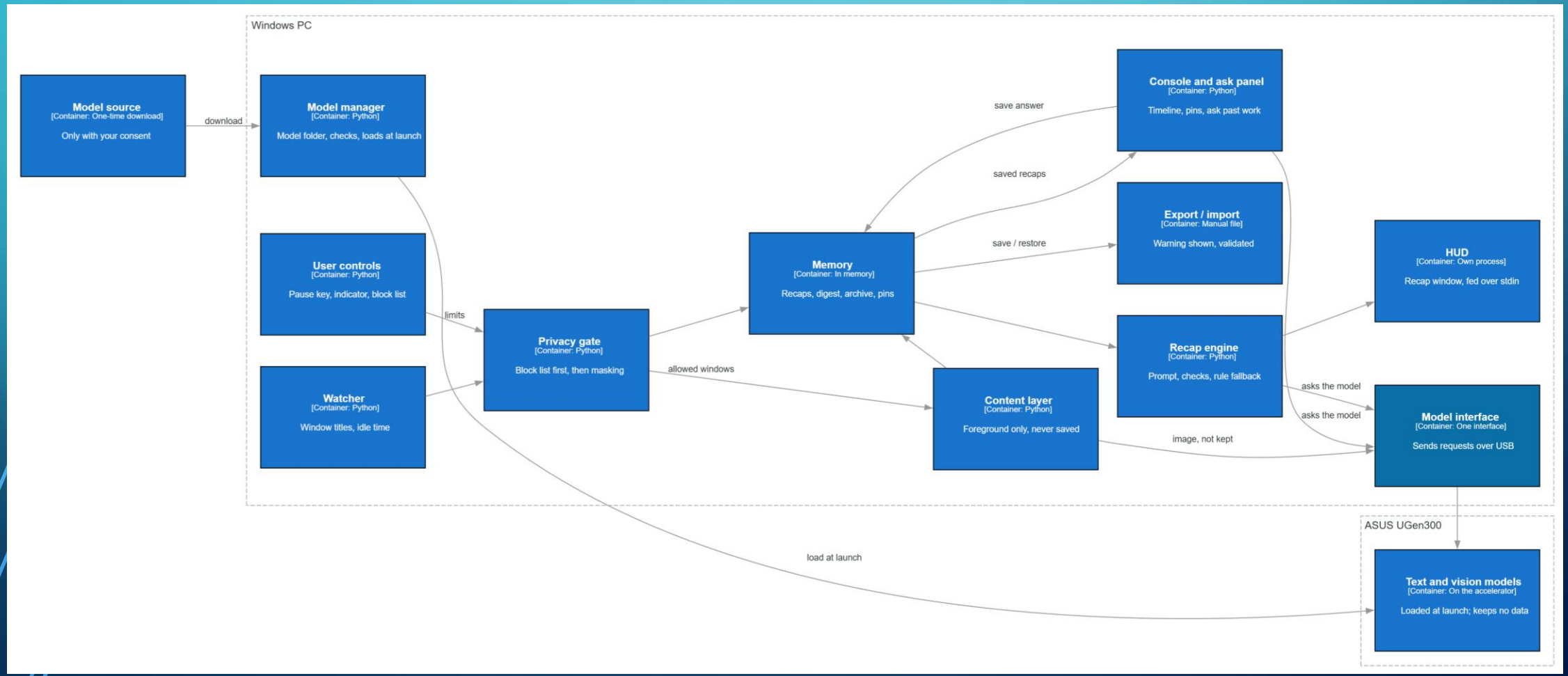
Architecture of Stage I

Here, everything runs on the laptop, and UGen300 is not connected yet.



Program Architecture

Target design: light work stays on the PC; all AI models run on the UGen300. The only network use is a one-time model download.



System status: built and target

Modules	Built (implemented)	Target (not built yet)
Sense	Window titles and idle time; AFK and return detection	Reading the window after 10 s idle and checking for changes
Protect	Block list, masking of emails, phones and card numbers; recap handed over standard input	**High priority:** pause key, capture indicator, editable block list. Network audit.
Remember	Last 25 window titles, in memory only	Pins with notes, 30-minute digest, 60-minute archive, saving with encryption, export and import
Recap	Rules write line 1; the model writes lines 2–3 and is checked; recap window at the bottom right	Vision description of the window; removing copied names
Ask		Ask panel; answers cite the saved recap they used
Hardware	Model interface with Ollama on the laptop	**High priority:** UGen300 backend with on-device measurements. First-launch model download.

High priority = built first in Stage II. The other targets follow as time allows.

Current limitationsand next step

Current limitations:

- Only English data is tested.
- UGen300 has not been used yet; its numbers come from the datasheet.
- A vision model can describe a screen wrongly, and it still copies some names into its description.
- GPU mode gave unusable vision output on this laptop.

Next (high priority): connect the UGen300, measure on the device, and add the pause key, capture indicator and block list. The other targets follow.

Expected Benefits

Who benefits

Users

Get back to work faster, without clicking through every window, and find earlier work by asking.

Companies

No cloud bills or extra per-use token fees; RecallPlug works offline, and sensitive work is designed to stay on their own computers.

How we will confirm:

We will confirm these benefits through hand-checked samples, timing on screen recordings, a network audit and on-device measurements in Stage II.

Targets

Measure	Goal	How we check
Recap window appears after you return	within 1.5 s	Timed on a screen recording
UGen300 text recap	within 10 s	The same benchmark script, run on the device
UGen300 vision description	within 15 s	The same script, on the device
Answer in the ask panel	within 10 s, citing the recap it used	Timed on the device; checked by hand on sample questions

Results Measured on a laptop, and the UGen300 datasheet.

MEASURED · laptop, 30 runs per scenario

Metric	Office	Web-heavy
Model call, median	0.18 s	0.15 s
Output speed	181 tokens/s	186 tokens/s
Host CPU, median	11%	9%
Host RAM	19.5%	19.5%
Tokens per recap	25	21.5

UGen300 datasheet

Model	Load	First token	Speed
Qwen2.5-1.5B (text)	5.05 s	0.37 s	6.82 tokens/s
Qwen2-VL-2B (vision)	6.23 s	0.93 s (image)	6.73 tokens/s

Datasheet figures are Hailo's benchmarks, not our measurements; on-device numbers come in Stage II. Without a GPU, one vision description took 84–93 s on this laptop's CPU (one laptop, stand-in model Qwen2.5-VL 3B, 5 screens): the heavy work the UGen300 takes over. The UGen300 outputs fewer tokens per second than this laptop's GPU; the recap is prepared before you return, so what matters is keeping the PC free.

Host: i7-12650H, RTX 4060 Laptop, 64 GB RAM, Windows 11; whole-computer averages.

Results

What we have run

Check	What we did	Result
Rules and the model	30 runs on the same fake windows at a low temperature. The model tends to continue the main document, so line 3 is checked as “grounded in a recent window” instead of “the last window”.	The strict check accepted the model once; the grounded check accepted it 30 times.
Vision check on 5 screens	5 hand-made screens, stand-in model Qwen2.5-VL 3B; and I check for it answer	5 out of 5 are correct and informative; some names are still copied, so we plan to stop names from appearing in its descriptions.

The pass rate shows format, not quality. GPU mode gave unusable output on this machine, so only the CPU-only run is reported.

Code, tests and benchmark scripts:github.com/geoffreysugs/recallplug

References and links

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3. Screenpipe. “What is screenpipe?” screenpi.pe/about

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5. [Datasheet] UGen300-USB-8G

Code: https://github.com/geoffreysugs/recallplug